

G16 Complete Graph Coloring

A report on the Greenwood-Gleason construction and Ramsey theory

ALGORITHM DESIGN

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1. Problem statement

The G16 coloring problem states: can the edges of K_{16} , a complete graph with 16 vertices where every pair is connected by an edge, be colored with three colors —Red, Green, and Blue so that no triangle has all its sides the same color?

A complete graph K_n has $n(n-1)/2$ edges. With $n = 16$, we have already added all 120 edges, so $C(16,3) = 560$ triangles to be checked. A coloring that satisfies the condition above is called a triangle-free 3-edge-colouring, and it has applications related to the Ramsey Number $R(3,3,3)$.

1.1 Connection to Ramsey Theory

The Ramsey number $R(3,3,3)$ is the smallest integer n such that any 3-coloring of K_n must contain a monochromatic triangle. Showing that K_{16} can be 3-colored *without* a monochromatic triangle proves $R(3,3,3) > 16$. Combined with the proof that every 3-coloring of K_{17} must contain one, we get $R(3,3,3) = 17$, a classical result established by Greenwood and Gleason in 1955.

2. Mathematical background

2.1 The finite field GF(16)

The construction is based on $GF(16)$, the finite field with 16 elements. So we express it as $GF(2)[x] / (x^4 + x + 1)$ In this field:

- The elements are polynomials of degree at most 3 over $GF(2)$ (actually bit strings), which can be simply represented as integers in $[0,15]$.
- Addition is bitwise XOR.
- Its multiplication is polynomial multiplication modulo the primitive polynomial $x^4 + x + 1$.
- The non-zero entries form a cyclic group of order 15 with respect to multiplication.

We select $\alpha = x = 2$ (binary representation: 0010) as a primitive element. Its powers create all 15 non-zero elements of $GF(16)$.

2.2 Partitioning non-zero elements

The 15 non-zero elements are classified in groups of 5 according to their discrete logarithm, modulo 3:

- G_1 (exponent $\equiv 0 \pmod 3$): $\alpha^0, \alpha^3, \alpha^6, \alpha^9, \alpha^{12} \rightarrow$ **Red** edges
- G_2 (exponent $\equiv 1 \pmod 3$): $\alpha^1, \alpha^4, \alpha^7, \alpha^{10}, \alpha^{13} \rightarrow$ **Green** edges
- G_3 (exponent $\equiv 2 \pmod 3$): $\alpha^2, \alpha^5, \alpha^8, \alpha^{11}, \alpha^{14} \rightarrow$ **Blue** edges

2.3 The coloring rule

For any two distinct vertices i and j in $\{0, 1, \dots, 15\}$, the color of edge $\{i, j\}$ is determined by:

$$\text{color}(\{i, j\}) = \log_\alpha(i \oplus j) \pmod 3$$

Where $i \oplus j$ is the $GF(16)$ difference (which equals the sum in characteristic 2), and $\log_\alpha$ gives the discrete logarithm base α . The result (0, 1, or 2) determines Red, Green, or Blue, respectively.

3. GF(16) power table

The table below shows all 15 non-zero elements of GF(16), their binary representations, and the color group they belong to:

Table 1: GF(16) elements and their color group assignments

i	α^i (decimal)	Binary (4-bit)	Exponent mod 3	Color Group
0	1	0001	0	Red
1	2	0010	1	Green
2	4	0100	2	Blue
3	8	1000	0	Red
4	3	0011	1	Green
5	6	0110	2	Blue
6	12	1100	0	Red
7	11	1011	1	Green
8	5	0101	2	Blue
9	10	1010	0	Red
10	7	0111	1	Green
11	14	1110	2	Blue
12	15	1111	0	Red
13	13	1101	1	Green
14	9	1001	2	Blue

Each color group contains exactly 5 elements. This balanced partition is the algebraic foundation of construction.

4. Why the construction works

4.1 The sum-free property

The main algebraic property is that each group G_k is sum-free: no two elements in G_k add (XOR) up to another element in G_k .

Let us assume for the sake of contradiction that there is a monochromatic triangle on vertices $\{x, y, z\}$ such that $[xy]$ and $[xz]$ have color k . Then:

$$\begin{aligned} x \oplus y &\in G_k \text{ (an edge } \{x, y\} \text{ has color } k) \\ y \oplus z &\in G_k \text{ (edge } \{y, z\} \text{ has color } k) \\ x \oplus z &\in G_k \text{ (edge } \{x, z\} \text{ has color } k) \end{aligned}$$

But $(x \oplus y) \oplus (y \oplus z) = x \oplus z$, meaning two elements of G_k sum to another element of G_k — a contradiction. Therefore, **no monochromatic triangle can exist**.

4.2 The Clebsch graph

Each color class forms an independent copy of the **Clebsch graph** — the unique **5-regular triangle-free graph on 16 vertices**. The three edge-disjoint Clebsch graphs partition all 120 edges of K_{16} perfectly (since each graph has 40 edges: $16 \times 5 / 2 = 40$).

Properties of each Clebsch graph (color class):

- 5-regular: every vertex has exactly 5 neighbors in this color
- Triangle-free: there is no monochromatic triangle in three vertices
- 40 edges: one-third exactly of the 120 edges of K_{16}
- Diameter 2: every pair of non-adjacent vertices share a neighbor
- Strongly regular: parameters $\text{srg}(16, 5, 0, 2)$

4.3 The Ramsey Number $R(3,3,3) = 17$

Since K_{16} admits a valid 3-coloring with no monochromatic triangle, we have $R(3,3,3) > 16$. Greenwood and Gleason also proved that every 3-coloring of K_{17} must contain a monochromatic triangle, giving $R(3,3,3) = 17$. This is an exact Ramsey number — rare in combinatorics.

5. Complete coloring of K_{16}

5.1 Neighbor lists

The table displays the five red neighbors (top two groups), as well as five green and five blue neighbors for each vertex. This confirms the 5-regular structure of each color class; every vertex has precisely 5 neighbors of each of the other two colors.

Table 2: Neighbor lists for all 16 vertices

v	Red neighbors	Green neighbors	Blue neighbors
0	1, 8, 10, 12, 15	2, 3, 7, 11, 13	4, 5, 6, 9, 14
1	0, 9, 11, 13, 14	2, 3, 6, 10, 12	4, 5, 7, 8, 15
2	3, 8, 10, 13, 14	0, 1, 5, 9, 15	4, 6, 7, 11, 12
3	2, 9, 11, 12, 15	0, 1, 4, 8, 14	5, 6, 7, 10, 13
4	5, 8, 11, 12, 14	3, 6, 7, 9, 15	0, 1, 2, 10, 13
5	4, 9, 10, 13, 15	2, 6, 7, 8, 14	0, 1, 3, 11, 12
6	7, 9, 10, 12, 14	1, 4, 5, 11, 13	0, 2, 3, 8, 15
7	6, 8, 11, 13, 15	0, 4, 5, 10, 12	1, 2, 3, 9, 14
8	0, 2, 4, 7, 9	3, 5, 10, 11, 15	1, 6, 12, 13, 14
9	1, 3, 5, 6, 8	2, 4, 10, 11, 14	0, 7, 12, 13, 15
10	0, 2, 5, 6, 11	1, 7, 8, 9, 13	3, 4, 12, 14, 15
11	1, 3, 4, 7, 10	0, 6, 8, 9, 12	2, 5, 13, 14, 15
12	0, 3, 4, 6, 13	1, 7, 11, 14, 15	2, 5, 8, 9, 10
13	1, 2, 5, 7, 12	0, 6, 10, 14, 15	3, 4, 8, 9, 11
14	1, 2, 4, 6, 15	3, 5, 9, 12, 13	0, 7, 8, 10, 11
15	0, 3, 5, 7, 14	2, 4, 8, 12, 13	1, 6, 9, 10, 11

5.2 Adjacency color mix

The full 16×16 adjacency matrix below shows the color of every edge. **R** = Red, **G** = Green, **B** = Blue, — = no edge (self).

Table 3: Full adjacency color matrix of K_{16}

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	-	R	G	G	B	B	B	G	R	B	R	G	R	G	B	R
1	R	-	G	G	B	B	G	B	B	R	G	R	G	R	R	B
2	G	G	-	R	B	G	B	B	R	G	R	B	B	R	R	G
3	G	G	R	-	G	B	B	B	G	R	B	R	R	B	G	R
4	B	B	B	G	-	R	G	G	R	G	B	R	R	B	R	G
5	B	B	G	B	R	-	G	G	G	R	R	B	B	R	G	R
6	B	G	B	B	G	G	-	R	B	R	R	G	R	G	R	B
7	G	B	B	B	G	G	R	-	R	B	G	R	G	R	B	R
8	R	B	R	G	R	G	B	R	-	R	G	G	B	B	B	G
9	B	R	G	R	G	R	R	B	R	-	G	G	B	B	G	B
10	R	G	R	B	B	R	R	G	G	G	-	R	B	G	B	B
11	G	R	B	R	R	B	G	R	G	G	R	-	G	B	B	B
12	R	G	B	R	R	B	R	G	B	B	B	G	-	R	G	G
13	G	R	R	B	B	R	G	R	B	B	G	B	R	-	G	G
14	B	R	R	G	R	G	R	B	B	G	B	B	G	G	-	R
15	R	B	G	R	G	R	B	R	G	B	B	B	G	G	R	-

6. Verification

The following table exhaustively verifies all required properties of the coloring:

Table 4: Verification of coloring properties

Check	Result	Status
Total edges in K_{16}	120	PASS
Red edges	40	PASS
Green edges	40	PASS
Blue edges	40	PASS
Monochromatic triangles	0	PASS
Vertices with $5R+5G+5B$ deg.	16 / 16	PASS
Each color class is 5-reg.	Yes	PASS

The verification confirms: **zero monochromatic triangles** across all $C(16,3) = 560$ triangles, perfectly balanced edge distribution (40 edges per color), and strict 5-regularity of each color class.

7. Algorithm summary

The full process of building the coloring:

1. Construct $GF(16)$: Map elements 0–15 to polynomials over $GF(2)$, multiplied mod $x^4 + x + 1$.
2. Choose a primitive element: Show that $\alpha = 2$ generates the complete field $GF(2^4)$ of 15 non-zero elements through multiplied recurrence.
3. Build the log table: For each non-zero element v , store $\log_alpha(v)$ = the smallest i such that $\alpha^i = v$.
4. Assign colors: For every edge $\{i, j\}$ where $i \neq j$, compute $d = i \text{ XOR } j$, look up $\log_alpha(d)$, and assign color = log mod 3.
5. Verify: Check all 560 triangles to confirm no monochromatic triangle exists.

Time complexity: $O(n^3)$ for the verification step (checking all triangles). The coloring assignment is $O(n^2)$. Space complexity is $O(n^2)$ for the adjacency matrix.

7.1 Java implementation

Two Java programs were developed to implement and visualize this construction:

- G16Coloring.java : console application to build $GF(16)$ tables, calculate full edge coloring, display neighbor lists / adjacency matrix, and exhaustively check the coloring with pass/fail report.
- G16ColoringGUI.java: SWING-based graphical program that allows you to render the K_{16} graph with edges colored according to their color class, select an individual vertex in order to view its neighbor list, enable/disable individual color classes, change edge opacity, and choose between 3 different layouts (circle, 2 rings Graph layout, and 4×4 grid).

8. Conclusion

The Greenwood-Gleason construction elegantly solves the G16 coloring problem using the algebraic structure of the finite field $GF(16)$. Mapping the 15 non-zero field elements to three color groups according to logarithm in base 3, and coloring each edge with the group of XOR of its endpoints, yields a valid 3-edge-coloring for K_{16} without a monochromatic triangle.

Therefore, it is a beautiful construction that reduces a combinatorial problem to an algebraic one: the coset of the cosets of the subgroup derived from cubes in $GF(16)^*$ sum-free property directly translate to triangle free graph. Each of the three color classes is a copy of the Clebsch graph, and together they form a perfect partition of K_{16} .

This result, combined with the upper bound proof, establishes $R(3,3,3) = 17$ — one of the few Ramsey numbers known exactly — and remains a landmark in combinatorics and graph theory.

9. References

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